MIL-D-36091
(BuMed)
2 August 1961

MILITARY SPECIFICATION

DISKS, MICROMETER, MICROSCOPE EYEPIECE

1. SCOPE

1.1 Scope. This specification covers micrometer disks (reticules), suitable for use in microscope eyepieces for purposes of microscopical measurements or counts.

1.1 Classification. Micrometer disks shall be of the following types, as specified (see 6.1):

Type I - Straight scale
Type II - Whipple

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up, Paperboard

Military

MIL-L-10547	Liners, Case, Waterproof

STANDARDS

Military

MIL - STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD -129	Marking for Shipment and Storage

FSC 6650

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The disks shall be manufactured from high-quality optical glass, having a refractive index of 1.5725 ± 0.0030, and shall be free of bubbles, striations, scratches, and digs.

3.2 Overall dimensions.

3.2.1 Diameter. Disks shall have an overall diameter of not less than 20.96 and not more than 21.20 millimeters (mm).

3.2.2 Thickness. Disks shall have an overall thickness of not less than 1.0 and not more than 2.0 mm.

3.3 Beveling. The disks shall be concentrically bevelled on both sides. Bevels shall average not less than 0.20 mm. nor more than 0.50 mm.

3.4 Parallelism. The top and bottom surfaces shall be parallel within 0° 10'.

3.5 Centricity. The center of the scale or grid area shall be within ± 0.2 mm. of the geometric center of the disk.

3.6 Graduations, grids, and numbers. All graduations, grids, and numbers shall be etched on the bottom side of the disk so as to read through the glass when positioned in the eyepiece.

3.6.1 Type I. The scale of the type I disks shall have the graduations, numbers, and dimensions as shown in figure 1.

3.6.2 Type II, The grid of the type II disks shall have the subdivisions and dimensions shown in figure 2 for the indicated size.

3.7 Etching. All ruled scales, areas, and numbers shall be permanent, accurate, and sharply cut. When placed and viewed through the eyepiece in a microscope, disk etchings shall present highly defined rulings.

3.8 Workmanship. Workmanship shall be first class throughout. Micrometer and counting disks shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

For visual examination	Inspection level	AQL (defects per hundred units)
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	L4	0.65

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4.2.2 For tests. Tests shall be performed on five sample disks from each lot in accordance with table I for the characteristics specified with no indication of failure. These characteristics are applicable to individual unit requirements.

Table I

Characteristic	Requirement	Test method
Refractive index	3.1	4.4.1
Beveling	3.3	4.4.2
Parallelism	3.4	4.4.3
Centricity	3.5	4.4.4
Graduations, grids, and numbers	3.6	4.4.5
Type I scale	3.6.1	4.4.6
Type II grid	3.6.2	4.4.7

4.3 Examination. The disks shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Glass not quality specified
102 Scale or grid not as specified

Minor

201 Glass not free of bubbles, striations, scratches and digs.
202 Beveling missing
203 Edge of disk chipped
204 Disk out of round

*Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The disks shall be examined for defects in dimensions. Any dimensions not within the tolerances specified herein shall be classified as a defect.

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4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Refractive index. The refractive index of the disks shall be determined by the use of a spectrometer.

4.4.2 Beveling. Place a calibrated micrometer scale disk in a microscope eyepiece, and using a high-power objective measure the beveling on four places on both sides of the disk. Average the readings obtained from each side.

4.4.3 Parallelism.

Apparatus:
Millimeter micrometer

Procedure; With the micrometer, measure the thickness of the disk just inside the bevel. Turn the disk approximately 180° and take another measurement The difference between the two measurements shall be not greater than 0.062 mm. Turn the disk approximately 90° from the point of the last measurement and take a measurement. Turn the disk approximately 180° and take another measurement. The difference between the two measurements shall be not greater than 0.062 mm.

4.4.4 Centricity of scale or grid.

Apparatus:
Monocular microscope
Cross hair eyepiece disk, the cross hairs of which are centered within ± 0.02 mm. of the geometric center of the disk.
Stage micrometer with 2-mm. scale divided into two hundred 0.01-mm. units, which has been mounted on a 25- by 75-mm slide.
Matched pair of Huygenian eyepieces.

4.4.4.1 Procedure for scale-type disk. Place the disk to be tested into one of the matched eyepieces, and the cross hair disk into the other. Place the eyepiece containing the cross hair disk into the microscope, and rotate until cross hairs are in a north-south, east-west position. Postion the stage micrometer so that the center line superimposes on the north-south cross hair. Remove the cross hair eyepiece and replace with the eyepiece containing the disk to be tested. If the center graduation of the micrometer disk does not superimpose on the stage micrometer center line, measure and record the distance, left or right, that the micrometer disk center line is off. Place the cross hair eyepiece in the microscope and position as above. Rotate the stage micrometer so that the center line is superimposed on the east-west cross hair. Remove the cross hair eyepiece and replace with the eyepiece containing the disk to be tested. Rotate the eyepiece so that the center line of the micrometer disk bisects the stage micrometer center line. Measure the distances from the stage micrometer center line to each end of the eyepiece micrometer center line. Compute the position of the scale relative to the geometric center of the disk.

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4.4.4.2 Procedure for grid-type disk. Place the cross hair and grid disk to be tested as specified in 4.4.4.1. Place the cross hair eyepiece in the microscope and position as specified in 4.4.4.1. Position the stage micrometer so that the center line superimposes on the north-south cross hair. Remove the cross hair eyepiece and replace with the eyepiece containing the disk to be tested. If the center graduation of the micrometer disk does not superimpose on the stage micrometer center line, measure and record the distance, left or right, that the grid disk center line is off. Rotate the grid disk 90° and proceed as above. Compute the position of the grid relative to the geometric center of the disk.

4.4.5 Graduations, grids and numbers.

Apparatus:
Microscope

4.4.5.1 For scale type. Place the micrometer disk into the eyepiece so that the etched scale side of the disk faces away from the top lens of the eyepiece. Place the eyepiece into the ocular tube of the microscope. Numbers shall read in consecutive order from left to right (see figure 1).

4.4.5.2 For grid type. Place the counting disk into the eyepiece so that the etched grid side of the disk faces away from the top lens of the eyepiece. Place the eyepiece into the ocular tube of the microscope. The smallest grid shall be in the lower left quadrant bounded on the north by the horizontal major cross line and on the east by the vertical major cross line (see figure 2).

4.4.6 Scale for type I.

Apparatus:
Microscope

Stage micrometer with 3-mm. scale divided into two hundred 0.01-mm. units, which has been mounted on a 25- by 75-mm. slide.

Procedure. Place the micrometer disk into the eyepiece and as described in 4.4.5.1. By means of the stage micrometer, measure for all scale dimensions shown in figure 1.

4.4.7 Grid for type II.

Apparatus:
Same as 4.4.6.

Procedure. Place the counting disk into the eyepiece as described in 4.4.5.2. By means of the stage micrometer, measure for all grid dimensions shown in figure 2.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) disk, of the type specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II - Packaging quantities

Type	Unit package	Intermediate package	Exterior container
I II	1 unit	12 units	144 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-566 or PPP-B-676. Carton shall be designed with suitable interior fittings, or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Each carton shall be adequately secured to prevent accidental opening.

5.1.3.2 Intermediate package.--Intermediate package shall be a carton of appropriate size and design, or a complete wrapping of kraft paper of suitable weight. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits fof the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-621, table III.

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5.2.1.2 Case liner. Each level A box shall be lined with a water-proof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be constructed in accordance with Federal Specification PPP-B-621, table I, or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or order.

5.3 Marking

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with 5.2 of Military Standard MIL-STD-129. In addition, contract number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in acoordance with Military Standard MIL-STD-129, except as follows: The letter "L" shall precede the Federal Stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification
(b) Type required (see 1.2)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

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6.3 Transportation description. Transportation description applicable to these items are:

Optical goods or instruments, NOIBN, in boxes

Carload minimum weight 10,000 pounds, subject to Rule 34 of the Uniform Freight Classification Series

Truckload minimum weight 10,000 pounds in accordance with National Motor Freight Classification (A Series)

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Air Force - AFCSG

Preparing Activity:

Navy - MMSA

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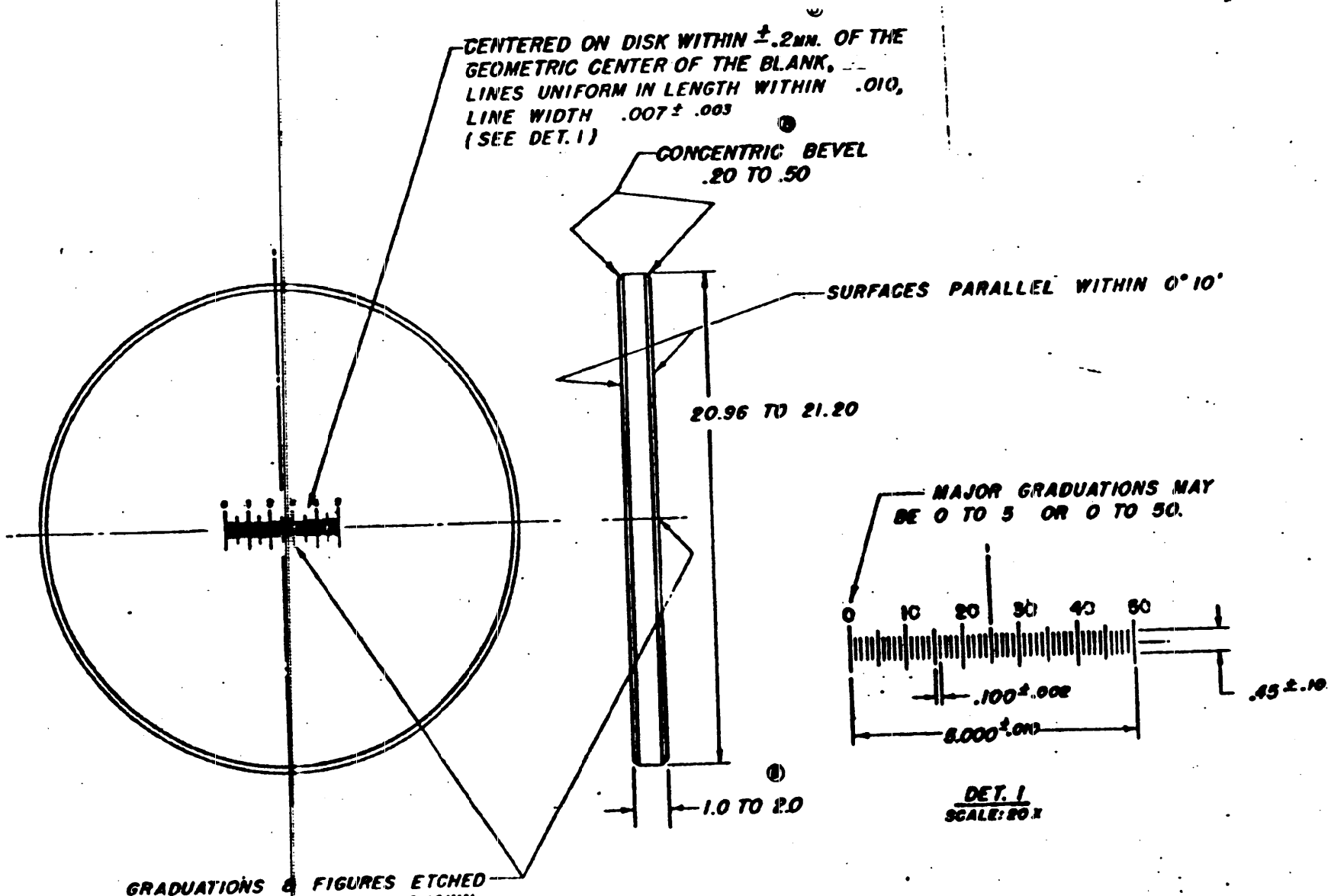
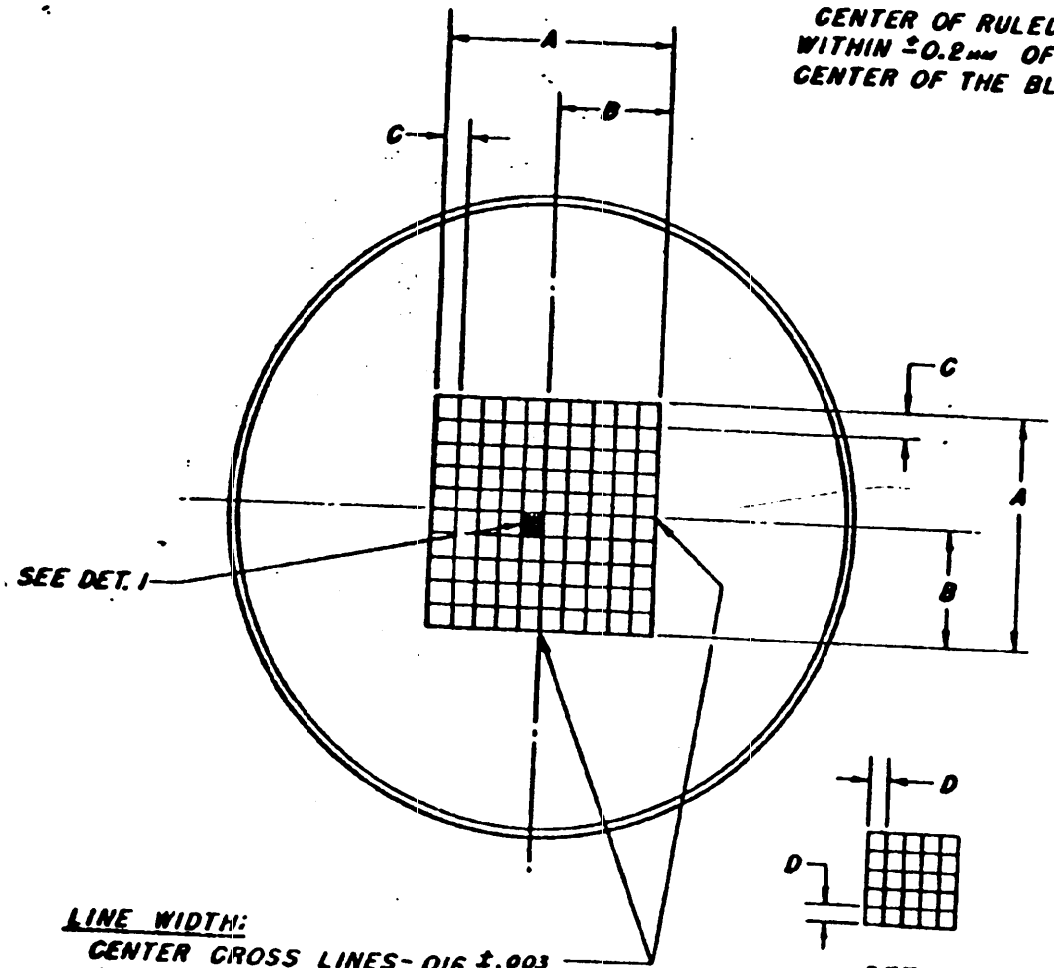


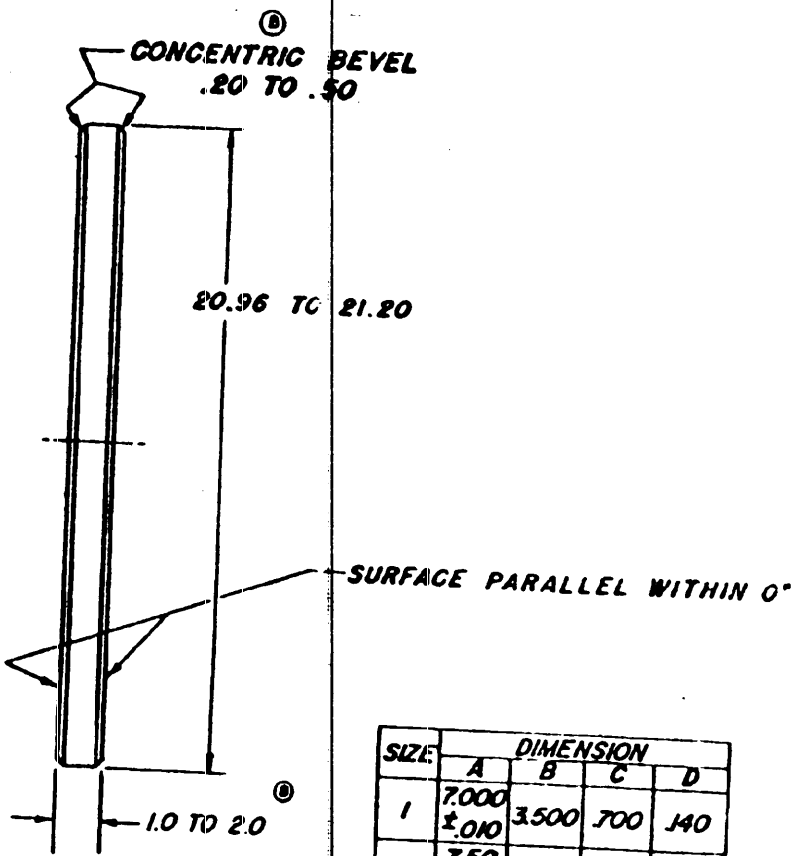
FIG. 1

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CENTER OF RULED AREA MUST BE WITHIN ±0.2mm OF THE GEOMETRIC CENTER OF THE BLANK.

DET. 1
SCALE: 32X



SIZE	DIMENSION			
	A	B	C	D
1	7.000 ±.010	3.500	.700	.140
2	7.50 ±.10	3.750	.750	.150
TOLERANCES		±.050	±.010	±.005

NOTE: DIMENSIONS ARE IN MILLIMETERS

LINE WIDTH:
CENTER CROSS LINES - .016 ±.003
OTHER LINES ——— .007 ±.003
LINES MAY EXTEND 0.1 BEYOND OUTER SQUARE.

FIG. 2

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